

Andres Villalobos

Senior Software Engineer

(512) 447-2893 | andres.villalobos@email.com | Austin, TX

EXPERIENCE

Software Engineer, BioData Dynamics | Austin, TX | March 2024 – Present

I work within the clinical data integration team on the PatientSync platform, a system that processes electronic health records from hospital networks across the Southwest region. My primary responsibility is maintaining and extending the ETL pipeline that ingests lab results, imaging metadata, and clinical notes from disparate EHR systems into our normalized database. I've contributed to the addition of a new LLM-based feature that extracts structured clinical findings from unstructured radiology reports, working closely with the ML team to integrate their model predictions into our existing data validation layer. The extraction workflow now runs daily across roughly 2,000 facilities, handling approximately 50,000 documents per day. I implemented several validation checks and error-handling logic to catch formatting edge cases that the initial model struggled with. Beyond the implementation, I participated in meetings with our hospital partners to understand how their staff interprets the extracted data, which helped surface issues with confidence scoring that we fed back to the ML team. The system has been in limited production for four months with positive feedback from clinical coordinators who use it to reduce manual data entry by an estimated 40%.

Junior Software Engineer, HealthConnect Solutions | San Antonio, TX | July 2023 – February 2024

I supported the backend development of MediFlow, a patient appointment and referral management system used by eight primary care networks in central Texas. My work focused on API integration endpoints that connected our system to third-party insurance verification services and pharmacy networks. I built and tested several REST API wrappers that standardized incoming data from partners with different API schemas, ensuring consistent error handling and retry logic. During my time here, I debugged production issues related to appointment scheduling conflicts and implemented fixes that reduced double-booking incidents by 30%. I also spent time in a few customer support calls, where I helped troubleshoot integration issues for two smaller clinic networks. The exposure to real clinical workflows and the constraints of healthcare data privacy taught me a lot about why certain architectural decisions matter. Before the role concluded, I documented the integration patterns I'd developed so that the team could reuse them for future partner onboarding.

Student Software Developer, University of Texas Medical School Computing Lab | Austin, TX | September 2022 – June 2023

I assisted faculty with internal web tools used for research data management and analysis. I contributed to a Django application that tracked longitudinal patient outcomes for a diabetes research cohort, writing features for data visualization and export functionality. This internship gave me early exposure to HIPAA compliance considerations and reinforced the importance of careful data handling in healthcare contexts.

TECHNICAL SKILLS

Python, JavaScript, SQL, REST APIs, Docker, AWS (RDS, Lambda), Git, Django, PostgreSQL, LLM prompt engineering, ETL pipeline patterns, healthcare data standards (HL7, FHIR basics)

EDUCATION

Bachelor of Science in Computer Science, University of Texas at Austin | May 2022